

Machine Learning Fundamentals

1. Supervised Learning

Supervised learning involves training a model on a labelled dataset. The goal is to predict an output (y) based on input features (x).

- **Linear Regression:** Used for predicting continuous numerical values, such as estimating the market price of a car.
- **Logistic Regression:** A classification algorithm used to assign data into categories, such as "Spam" or "Not Spam."
- **Decision Trees:** Uses a flowchart-like structure to make decisions based on data attributes.

2. Unsupervised Learning

This method uses unlabelled data to find hidden structures or patterns without explicit instructions.

- **K-Means Clustering:** Groups similar data points together. It is commonly used for customer segmentation in marketing.
- **PCA (Principal Component Analysis):** Reduces the number of variables in a dataset while keeping the most important information.

3. Reinforcement Learning

An agent learns by interacting with an environment. It receives "rewards" for correct actions and "penalties" for mistakes.

- **Use Case:** Training autonomous robots to navigate a warehouse or teaching an AI to play chess at a grandmaster level.